Multi-Horizon Joint-Embedding Predictive Architectures and Coordinate Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

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***Abstract*—Monitoring systems such as spacecraft and industrial plants must flag sensor faults quickly while staying quiet during normal operation. Detectors that reconstruct raw signals fire often on harmless jitter and drift. Joint-Embedding Predictive Architectures (JEPAs) instead predict compact learned summaries and raise far fewer false alarms, yet single-score JEPAs sum every coordinate into one number, letting drifting coordinates inflate the threshold and hide faults elsewhere. We replace that scalar score with per-coordinate circuit breakers: each coordinate's error across future horizons receives its own healthy-calibrated threshold, and an alarm fires when a quorum trips. We study Laminar-JEPA-Lite, a 128-breaker operational architecture, and Laminar-JEPA, a 224-breaker ablation reference. Primary evaluation uses 118 untouched archive streams across nine datasets, separating model fitting, checkpoint selection, adaptation, and calibration onto disjoint healthy blocks. Under buffered attribution, Laminar-JEPA-Lite raises event recall from 59.0 to 67.4 percent and Point-F1 from 0.143 to 0.156 over a single-score multi-horizon JEPA while lowering held-out false-positive rates from 19.5 to 14.3 percent. Under live causal reporting, it raises event recall from 52.5 to 63.8 percent across all nine datasets, halves delay, and beats system-matched coordinate aggregators; pooled causal Point-F1 rises modestly from 0.076 to 0.079 (doubling from 0.040 to 0.079 when calibration reaches 500 windows), while causal held-out false-positive rates are right-skewed (0.78 percent median, 7.0 percent mean from short calibration histories). Design-stage streams show 1 versus 22 to 24 percent false-positive rates against reconstructors; slow ramps remain undetected.**

***Index Terms*—Anomaly detection, Multivariate time series, Representation learning, Self-supervised learning, Statistical process control.**

I. Introduction

Spacecraft, power plants, and industrial machines stream readings from dozens of sensors at once [1]–[4]. A detector watching this telemetry has two jobs: raise an alarm soon after something breaks, and stay quiet the rest of the time. Faults are rare and seldom labeled, so the detector must learn what normal looks like from healthy history alone. Operators lose trust quickly when a monitor alarms several times a day on healthy data, and an ignored monitor catches nothing.

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The most common approach trains a network to reproduce raw sensor signals and raises an alarm when the reproduction is poor; TimesNet [5], TranAD [6], and Anomaly Transformer [7] follow this recipe. In practice, much of raw telemetry cannot be copied at all. Random sensor jitter leaves an irreducible error, and benign thermal drift or duty-cycle changes degrade the copy even when nothing is broken. Summing errors across 25–55 channels measured in different units allows one noisy channel to dominate, producing heavy-tailed spikes on healthy data.

These spikes defeat threshold calibration. Setting thresholds on healthy windows with Streaming Peaks-Over-Threshold (SPOT) [8] still yields 22%–24% causal false-positive rates (FPR) for reconstruction detectors on later healthy data across our 19 design-stage development streams, roughly one step in four. Because faults span many steps, alarming at random 24% of the time already hits 60.0% of fault episodes; TimesNet catches 58.8%, reflecting frequent alarms rather than selective fault recognition.

A Joint-Embedding Predictive Architecture (JEPA) [9], [10] avoids copying raw signals. An encoder compresses each window into a 32-number summary, and a predictor forecasts the summary of the next window. Discarding unpredictable jitter yields lighter error tails and cuts healthy false alarms on those development streams by an order of magnitude (Contribution 3).

Standard JEPA detectors, however, still collapse the 32 prediction errors into one scalar score and one threshold. When one or two summary features drift slightly as a machine warms up or switches load, their growing errors push the shared threshold upward. A fault that shifts the remaining 30 features moderately then stays below that raised threshold and goes missed. We call these drifting features *hijacker dials*. On our development streams, single-score JEPAs reach causal Point-F1 (step-level alarm accuracy) of only 0.050–0.055 and need 138–154 steps to flag a fault.

Two evaluation shortcuts also inflate published numbers: point adjustment credits an entire fault segment if one step inside it is flagged [11], and thresholds tuned with test labels or calibrated on adaptation data understate false alarms. We score every step without point adjustment, fit thresholds on healthy data only, separate model fitting, checkpoint selection, adaptation, and calibration into disjoint healthy blocks on our primary archive, and reserve 118 streams never inspected during design.

We replace the single scalar alarm with a panel of circuit

breakers, like a building breaker box where a short in one circuit trips its own switch regardless of total load. Across target prefixes of 16, 32, and 64 steps, each of the 32 coordinates receives separate *dials* tracking horizon error size and persistent drift, each with its own healthy-calibrated threshold, and the system alarms when a quorum of dials trips together. Drifting dials widen only their own thresholds instead of blinding the rest of the panel. Targeting abrupt to medium-onset faults (within roughly one 256-step context window), we study two variants (Fig. 1):

1) *Laminar-JEPA-Lite*, our preferred operational architecture, predicts three horizons in parallel behind one layer-normalization (LayerNorm) step [12] and monitors $4 \times 32 = 128$ position-plus-drift dials under one quorum (35%–52% fewer parameters and $2.43\times$ faster inference).
2) *Laminar-JEPA*, a 224-dial ablation reference variant, scores four views of each coordinate's error plus 32 onset relays and optional weight editing to isolate which views aid live causal versus buffered attribution; more dials do not imply a better operational detector.

A. Contributions

Thirty-three development streams (where adaptation and thresholds share calibration data) provide exploratory design-stage evidence, while 118 untouched archive streams across 9 datasets (of 141 streams across 12 datasets) under a disjoint five-block protocol provide primary evidence of generalization and calibration (Section V-B). We report live *causal* streaming separately from *buffered* after-the-fact attribution:

1) *Event coverage, buffered accuracy, and archive false-alarm behavior:* On the 118 untouched streams, *Laminar-JEPA-Lite* raises buffered event recall (ER) from 59.0% to 67.4% and buffered Point-F1 from 0.143 to 0.156 ($p = 0.022$) over a single-score multi-horizon JEPA while lowering held-out buffered FPR ($19.5\% \rightarrow 14.3\%$), retaining a $+10.9$-point recall lead at a matched 5% test FPR. Under live causal reporting, it raises ER from 52.5% to 63.8% across all 9 datasets (sign-flip $p = 0.0039$; all 12 leave-one-dataset-out (LODO) folds), halves delay (70.8 vs. 133.5 steps), and beats system-matched coordinate aggregators. Pooled causal Point-F1 rises modestly from 0.076 to 0.079 (95% stream confidence interval (CI) $[-0.019, +0.023]$, $p = 0.014$), doubling from 0.040 to 0.079 ($p = 2.7 \times 10^{-5}$) on adequately calibrated streams ($N_{\text{cal}} \geq 500$, $n = 59$). Archive causal held-out FPR is right-skewed: median 0.78% (0.61% for $N_{\text{cal}} \geq 500$), 87.3% of streams $\leq 5\%$, and mean 7.0% driven by short calibration histories.
2) *Contributing factors:* Design-stage matched-backbone ablations indicate that LayerNorm and barycenter centering contribute to lighter tails and less drift under these protocols, whereas spectral reweighting has a null effect.
3) *Design-stage comparison with reconstructors:* On 19 development streams, latent prediction holds causal FPR at 0.6%–2.4% versus 22.2%–24.3% for reconstructors, and at a matched 1% test FPR *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% and 56.0% of events versus 21.7% for TimesNet.
4) *Explicit limits:* Proposition 1 gives geometric bounds whose scalar score reweighting is null; autocorrelation limits extreme-quantile precision on short calibration blocks; an optional channel branch lifts recall on synthetic single-sensor faults (80.7%–96.5%) only by raising test FPR from 1.5% to 6.8%–19.4%; and slow ramps ($L \geq 4C$) and natural sensor failures remain out of reach.

II. Related Work

Observation-Space Forecasting and Reconstructive Detectors: Deep detectors typically forecast or reconstruct raw windows and threshold $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$: autoregressive long short-term memory (LSTM) forecasters [4] model spacecraft channels directly; TranAD [6] pairs adversarial training with self-conditioning attention; Anomaly Transformer [7] contrasts Gaussian prior and series-level attention; and TimesNet [5] reshapes 1D series along Fast Fourier Transform (FFT) periods for 2D convolution. Equal channel-wise reconstruction losses fit stochastic jitter, so summing errors across $K \geq 25$ channels yields heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) spikes and high false alarms under baseline shift [13].

Contrastive and Joint-Embedding Predictive Architectures: Representation methods encode windows into compact latents [14]. Contrastive approaches—temporal-contextual contrasting [15], TS2Vec [16], Neural Contextual Anomaly Detection (NCAD) [10] with synthetic anomalies, and dual-attention DCdetector [17]—depend on whether synthetic perturbations or patch discrepancies match test faults (Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [18], [19] predict target embeddings z_{tgt} from context z_{ctx} without observation decoding or negative pairs, avoiding collapse via exponential moving average (EMA) targets [20] or variance-covariance terms [21]–[23].

Residual Decomposition, Conformal Thresholding, and Statistical Process Control: Telemetry reflects underlying system dynamics [24]. Neural ordinary differential equations (ODEs) [25], Physics-Informed Neural Networks (PINNs) [26], Hamiltonian/Lagrangian nets [27], [28], and Koopman lifts [29]–[33] model structured dynamics, though partial differential equation (PDE) penalties require known laws and exhibit stiff gradients on uncalibrated sensors [34]; we instead borrow the algebraic mean-plus-fluctuation identity of Reynolds averaging [35] strictly as a multi-horizon residual summary. Multivariate statistical process control (SPC) charts such as Hotelling's T^2 [36] and Multivariate Exponentially Weighted Moving Average (MEWMA) [37] collapse whitened coordinates into one scalar, as do 1D deep scores calibrated via extreme-value theory [38] and Generalized Pareto Distribution (GPD) tails in SPOT [8] under serial dependence [39]. Per-coordinate alternatives such as univariate Cumulative Sum (CUSUM) banks [40] and per-coordinate split-conformal rules under Bonferroni, Šidák, or Benjamini–Hochberg (BH) false-discovery-rate (FDR) control [41] hit a discrete calibration floor $(N_{\text{cal}} + 1)^{-1}$ on short histories (Section VI-A). All methods are evaluated without point adjustment [11], [13], [42].

III. Multi-Sensor Telemetry Benchmark Suite

Each test case in this paper is a *stream*: one recording from one system, with K sensor channels sampled over time. Every stream has a healthy (nominal) history, used for training and calibration, followed by a labeled test segment that contains the faults. We use two collections of streams. A 45-stream development suite drawn from 11 datasets guided every design decision and supplies the main development benchmark. A larger 141-stream archive drawn from 12 datasets supplies the final evaluation, and 118 of its streams were never inspected during design.

Development suite: The 45 streams come from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [43], [44], TimeEval [13], and mTSBench [45]. They cover three kinds of systems with channel counts from 2 to 72 and very different fault types:

1) *Spacecraft and solar telemetry:* six SMAP streams (K=25), six MSL rover streams (K=55), and SWAN-SF solar magnetograms [46] (K=38). NASA labels each SMAP and MSL fault as either an out-of-range *point* fault, where a value leaves its normal range, or an in-range *contextual* fault, where values stay in range but behave abnormally together [4] (Table V(a)).
2) *Wearable motion sensing:* six Daphnet Freezing-of-Gait (FoG) streams [43] recorded at 64 Hz (K=9, 72 episodes), so the S=64-step target window spans one second.
3) *Industrial processes and infrastructure:* 14 Gasoil Heating Loop (GHL) streams (K=16) whose 26 faults are slow ramps, the Genesis pick-and-place testbed (K=18), GECCO water chemistry (K=9), Metro I-94 traffic (K=5), Room-Occupancy sensing [47] (K=5), CalIt2 building flow [44] (K=2), and six CICIDS network streams (K=72).

Thirty-three of these streams form the development benchmark: 19 non-GHL streams and the 14 GHL streams, spanning 7 datasets. Inclusion required enough healthy data for calibration ($N_{\text{cal}} \geq 500$ windows), at most 55 channels, and at most 1.2×10^5 test steps (Table IV(c)). The other 12 streams are reserved: five have very short healthy histories (764–1,492 steps; CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), the six CICIDS streams exceed the channel limit, and GECCO is evaluated in the archive instead.

Evaluation archive: The archive contains every qualifying stream from 12 datasets, requiring at least 100 calibration windows and at least one labeled anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, and 13 OPPORTUNITY streams, plus single streams from GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN. Healthy histories range from 1,920 to 132,481 steps and test segments from 2,158 to 114,001 steps. Removing the 23 streams used to tune the method leaves 118 untouched streams from 9 datasets (Section V-B).

How each stream is split: Every healthy history is divided in time order. The first 65% fits the model, and its first part also sets the per-channel standardization ($\boldsymbol{\sigma}$ floored at 10^{-2}). The next 17.5% calibrates alarm thresholds ($\mathbf{X}_{\text{cal}}$), and after a 64-step gap the last 17.5% measures false alarms on healthy data the thresholds have never seen ($\mathbf{X}_{\text{hold}}$). Archive runs further divide the model-fitting block (Section V-A).

Caveats: SMAP and MSL contain documented labeling flaws [42], and streams from the same archive often share hardware, so they are not fully independent. We therefore also treat whole datasets as units of analysis (7 for the development benchmark, 9 for the untouched streams, 12 for the full archive), using Dataset-Macro means, cluster-bootstrap confidence intervals (CI), exact dataset sign-flip permutation tests, and leave-one-dataset-out audits (Section V-A).

IV. Methodology: Laminar-JEPA and Laminar-JEPA-Lite

Fig. 1 shows the pipeline from raw telemetry $\mathbf{X} \in \mathbb{R}^{T\times K}$ (K channels over T steps) to alarms. In plain terms, it has four stages: (1) encode each window of recent data into a short summary vector; (2) predict the summaries of the next few stretches of data; (3) turn the prediction errors into a set of *dials*, each measuring one aspect of the error; and (4) give every dial its own threshold and raise an alarm when enough dials trip together. The networks are trained on healthy data only; stabilization weights and alarm thresholds are then fitted on later healthy blocks and frozen.

A. Stage 1: Multiscale Encoder and Barycenter Centering

A sliding window of length $W = C + S = 320$ advances one step at a time. At each position t it splits into a *context window* $\mathbf{x}_{\text{ctx}}$ of the past $C = 256$ steps and a *target window* $\mathbf{x}_{\text{tgt}}$ of the next $S = 64$ steps, which ends at the decision time $\tau = t+S-1$. Telemetry mixes slow trends with fast vibration, so the encoder first separates time scales. Multichannel inputs ($K > 1$) pass through a causal per-channel convolution and are split into four smooth frequency bands (dyadic shells of a 256-point real fast Fourier transform (rFFT), forming an amplitude partition of unity). Each band has its own causal convolution (kernels $\{15, 9, 5, 3\}$), and a softmax router feeds the bands into a causal dilated temporal convolutional network (TCN) [48]. Single-channel inputs skip the band split.

The encoders follow the Joint-Embedding Predictive Architecture (JEPA) [9]: instead of reconstructing raw waveforms, an online encoder E_θ and a target encoder E_ϕ map context and target windows to $D = 32$-dimensional summaries, each normalized by LayerNorm to length close to $\sqrt{D}$. The target encoder is a slowly updated copy of the online one, an exponential moving average (EMA) with momentum $m = 0.996$ (0.995 on archive cohorts) [18], [20]. Two problems arise in practice. Deep dilated TCNs blur brief spikes at fault onset, and changes in operating regime shift all summaries together. Two encoder components address them:

1) *Multi-depth tapping (keeps fast transients):* Each encoder also reads three intermediate feature maps ($d_{\text{model}} = 48$): the spectral stem, a middle TCN block, and the final TCN block. For each tap, the last time step and the root-mean-square (RMS) energy over time are projected to $\mathbb{R}^D$ by a two-layer GELU head and blended with the deep TCN output through a bounded gate ($g \leq 0.5$).

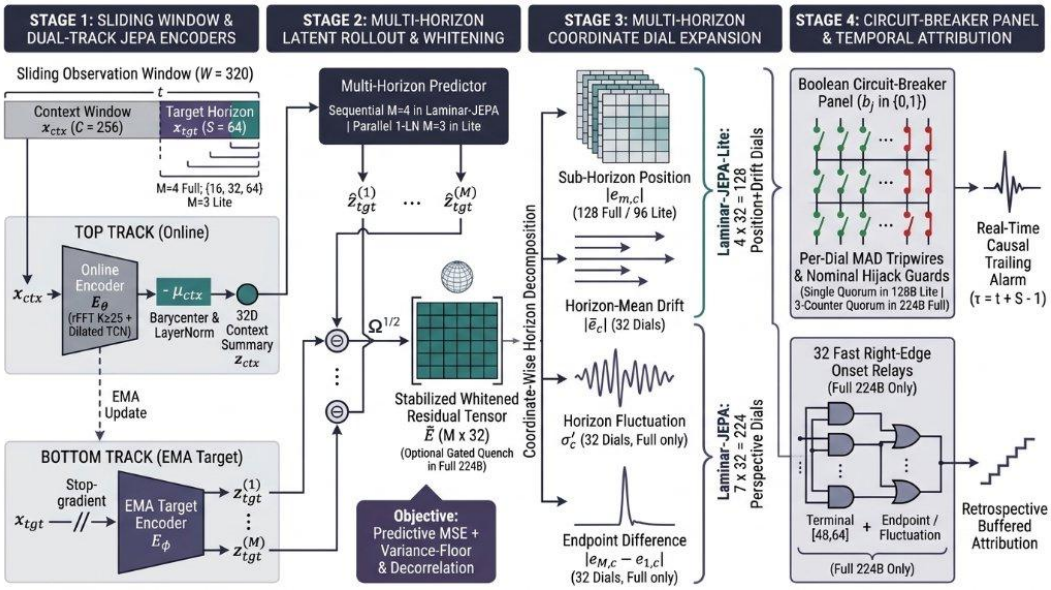


Fig. 1. Overview of the four-stage *Laminar-JEPA* and compact *Laminar-JEPA-Lite* pipeline: **(1)** sliding window (C=256, $S \in \{1..64\}$) and dual encoders (E_θ, E_ϕ) with conditional rFFT splitting ($K \geq 25$ in Lite), barycenter centering, and LayerNorm; **(2)** multi-horizon rollout (M=4 sequential in full *Laminar-JEPA*; M=3 parallel single-LayerNorm in *Lite*) under variance-floor and decorrelation loss plus resolvent whitening into $\tilde{\mathbf{E}} \in \mathbb{R}^{M\times 32}$; **(3)** horizon expansion into $7 \times 32 = 224$ dials (full) or $4 \times 32 = 128$ position-plus-drift dials (*Lite*); and **(4)** per-dial Boolean circuit breakers emitting causal (τ=t+S−1) or prefix-buffered alarms.

2) *Barycenter centering (removes common offsets):* During training, a running mean of the context summaries $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ (update rate 0.05, shared with E_ϕ) is subtracted from both context and target summaries before the final LayerNorm. Centering removes offsets learned in training; it does not guarantee invariance to unseen drift.

B. Stage 2: Multi-Horizon Prediction

The simplest JEPA detector, the single-horizon baseline *TS-JEPA*, uses a two-layer multi-layer perceptron (MLP) to predict one summary $\hat{z}_{\text{tgt}}$ of the whole 64-step target window, trained with predictive mean-squared error (MSE) and a VICReg-style log-determinant penalty ($\beta = 0.05$) [21]. Summarizing all 64 steps in one vector can average away a short fault. *Laminar-JEPA* therefore predicts how the target unfolds: it summarizes $M = 4$ nested prefixes of the target window, the first $S_m \in \{16, 32, 48, 64\}$ steps. Starting from z_{ctx} plus learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, a recurrent predictor P_{sub} produces $\hat{z}_{\text{tgt}}^{(1)}, \ldots, \hat{z}_{\text{tgt}}^{(M-1)}$ in sequence, and an anchor head P_θ predicts the full-window summary $\hat{z}_{\text{tgt}}^{(M)}$. The loss gives the full window weight $w_M = 1/2$ and each shorter prefix $w_m = 1/(2(M-1))$ (1/6 in the full model, 1/4 in Lite), and adds a penalty $\boldsymbol{\Delta}_\gamma$ that keeps the summaries from collapsing to a constant:

$$\mathcal{L}_{\text{Lam}} = \sum_{m=1}^{M} \frac{w_m}{D} \Big(\big\| \hat{z}_{\text{tgt}}^{(m)} - \text{sg}\big(z_{\text{tgt}}^{(m)}\big) \big\|_2^2 + \big\| \boldsymbol{\Delta}_\gamma\big(\hat{z}_{\text{tgt}}^{(m)}\big) \big\|_F^2 \Big), \tag{1}$$

where $\text{sg}(\cdot)$ stops gradients through E_ϕ and the penalty acts on the predicted summaries of each batch. Log-determinant penalties need batches larger than D; $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ works on small batches. Its diagonal entries $\sqrt{\max(0, \gamma - \sigma_d)}$ push each coordinate's standard deviation above $\gamma = 0.75$ (σ_d from biased batch variance plus 10^{-4}) and its off-diagonals $\Sigma_{dj}/\sqrt{2}$ penalize sample covariance (divisor $B-1$, both symmetric entries counted), so $\frac{1}{D}\|\boldsymbol{\Delta}_\gamma\|_F^2 = \frac{1}{D}\sum_{d=1}^{D} \max(0, \gamma - \sigma_d) + \frac{1}{2D}\sum_{d\neq j} \Sigma_{dj}^2$. At $B = 16 < D = 32$, this lifts target effective rank $\text{RankMe}(z_{\text{tgt}})$ [49] from 11.6 to 19.8.

C. Stage 3: Residual Whitening and Dial Expansion

Stage 3 turns prediction errors into dials. The signed errors $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ for the $M = 4$ horizons form a 4×32 error matrix $\mathbf{E}(t)$. Its entries have different natural sizes and are correlated, so they are first *whitened*: rescaled to unit variance and decorrelated using statistics from healthy training windows. We flatten $\mathbf{E}(t)$ into a vector of length MD (128 in the full model, 96 in Lite), estimate its mean and unbiased covariance on the $\mathbf{X}_{\text{fit}}$ windows, and with $\hat{\boldsymbol{\Sigma}}_{\text{res}} = \mathbf{U}\,\text{diag}(\lambda_k)\mathbf{U}^\top$ ($\mathbf{U}, \boldsymbol{\Omega} \in \mathbb{R}^{MD\times MD}$) use the damped precision matrix $\boldsymbol{\Omega} = \mathbf{U}\,\text{diag}\big(\lambda_k/(\lambda_k + 10^{-3})^2\big)\mathbf{U}^\top$. Multiplying centered errors by $\boldsymbol{\Omega}^{1/2}$ gives whitened errors $e_{m,c}^{\text{white}}(t)$; the damping constant 10^{-3}, fixed across all cohorts, shrinks directions with almost no training variance instead of amplifying their noise.

1) *Stabilization on a separate healthy block $\mathbf{X}_q$:* A coordinate whose error grows between training and later healthy data would be inflated by whitening. We detect such coordinates on a separate healthy block $\mathbf{X}_q$, never on the block $\mathbf{X}_{\text{cal}}$ that later sets alarm thresholds (Section V-A), and blend them toward a simpler per-coordinate scaling, the unwhitened error divided by its root-mean-square on $\mathbf{X}_q$, $\hat{e}_{m,c}^{\text{raw}}(t) = e_{m,c}(t)/\text{RMS}_q(e_{m,c})$. The blend starts when the whitened RMS on $\mathbf{X}_q$ exceeds 1.10, and all weights are frozen before $\mathbf{X}_{\text{cal}}$ is used:

$$\tilde{e}_{m,c}(t) = w_{m,c}\, e_{m,c}^{\text{white}}(t) + (1 - w_{m,c})\, \hat{e}_{m,c}^{\text{raw}}(t),$$
$$w_{m,c} = \frac{1}{1 + 2\max\big(0, \text{RMS}_q^{\text{white}}(m,c) - 1.10\big)^2}. \tag{2}$$

The full *Laminar-JEPA* can also run an optional gradient-free *weight-space quench*, which edits the few predictor

outputs that drift most on healthy data and keeps the edit only if false alarms fall. It uses only $\mathbf{X}_{\text{fit}}$ and $\mathbf{X}_q$ (legacy fixed-epoch development runs without $\mathbf{X}_q$ use $\mathbf{X}_{\text{cal}}$; Table I). Coordinates are ranked by a median-absolute-deviation (MAD) standardized instability score that combines mean and scale shifts from $\mathbf{X}_{\text{fit}}$ to $\mathbf{X}_q$, drift between the halves of $\mathbf{X}_q$, and residual autocorrelation, with a negative term for isolated spikes. For the top $k \in \{4, 2\}$ coordinates, four steps run in order. *(Q1)* Damp the selected predictor output rows and LayerNorm gains and shift $\mathbf{p}_m$ toward the $\mathbf{X}_q$ means; through the shared 32D LayerNorm this also moves unselected coordinates (Section VI-A). *(Q2)* Move residual centers 75% of the way toward the post-edit $\mathbf{X}_q$ means. *(Q3)* Scale off-diagonal precision couplings by 0.15 and rescale whitened $\mathbf{X}_q$ RMS. *(Q4)* Keep the edit (preferring $k = 2$ on ties) only if the $\mathbf{X}_q$ half-split FPR improves while at least 92% of the pre-edit quorum thresholds are preserved; then freeze all parameters.

2) *Expansion into $7D = 224$ dials:* For each of the 32 coordinates, the four horizon errors describe how the prediction error evolves across the target window. Borrowing the mean-plus-fluctuation split of Reynolds averaging [35] purely as algebra, we split them into a horizon mean $\bar{e}_c(t) = M^{-1}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)$ and a zero-mean fluctuation $F_{m,c}(t) = \tilde{e}_{m,c}(t) - \bar{e}_c(t)$. Four views of each coordinate give $7D = 224$ nonnegative dials $\mathbf{d}(t) \in \mathbb{R}^{224}_{\geq 0}$: (a) $4D = 128$ *sub-horizon positions* $|\tilde{e}_{m,c}(t)|$, the error size at each horizon; (b) $D = 32$ *horizon-mean drifts* $|\bar{e}_c(t)|$, a persistent offset shared by all horizons; (c) $D = 32$ *horizon fluctuations* $d_{\text{fluc},c}(t) = \sqrt{M^{-1}\sum_{m=1}^{M} F_{m,c}^2(t)}$, how much the error changes across horizons; and (d) $D = 32$ *endpoint differences* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$, error that appears late in the window, as at a fault onset.

The proposition below states two exact properties of this split. Part (i) says that drift and fluctuation measure non-overlapping parts of the position energy, and that the endpoint dial cannot exceed a multiple of the fluctuation dial. Part (ii) says the fluctuation pattern lives in at most $M - 1$ directions and defines R_{LT}, which is 1 when one pattern dominates and $1/(M-1)$ when the energy is spread evenly. These are geometric facts about the representation, not accuracy guarantees.

Proposition 1 (Horizon Energy Split and Spectral Concentration Bounds)**.** *For $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) with horizon mean $\bar{\mathbf{e}}(t)$ and fluctuation matrix $\mathbf{F}(t) = \tilde{\mathbf{E}}(t) - \mathbf{1}_M \bar{\mathbf{e}}(t)^\top \neq \mathbf{0}$: (i) for every coordinate c, $M^{-1}\sum_{m=1}^{M} |\tilde{e}_{m,c}(t)|^2 = |\bar{e}_c(t)|^2 + d_{\text{fluc},c}^2(t)$ and $d_{\text{accel},c}(t) \leq \sqrt{2M}\, d_{\text{fluc},c}(t)$; and (ii) the $M \times M$ horizon Gram matrix $\mathbf{G}(t) = \mathbf{F}(t)\mathbf{F}(t)^\top \in \mathcal{S}_+^M$ has ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ whose scale-invariant dominant-mode share for any $p > 0$ [50] satisfies*

$$R_{LT}\big(\mathbf{F}(t)\big) = \frac{\lambda_M^p}{\sum_{m=1}^{M} \lambda_m^p} \in \Big[\frac{1}{M-1},\, 1\Big], \tag{3}$$

reaching $\frac{1}{M-1}$ iff energy is isotropic across all $M-1$ active modes ($\lambda_2 = \cdots = \lambda_M > 0$, requiring $D \geq M-1$) and 1 iff $\mathbf{F}(t)$ has rank one. Empirically, scalar reweighting by R_{LT} has a null effect (Section VI).

Proof. Zero column sums ($\sum_{m=1}^{M} F_{m,c} = 0$) cancel the cross term in $M^{-1}\sum_{m=1}^{M}(\bar{e}_c + F_{m,c})^2$, and $(F_{M,c} - F_{1,c})^2 \leq 2(F_{M,c}^2 + F_{1,c}^2) \leq 2M\, d_{\text{fluc},c}^2$ gives (i). For (ii), $\mathbf{F}^\top \mathbf{1}_M = \mathbf{0}_D$ implies $\mathbf{G}\mathbf{1}_M = \mathbf{0}_M$ ($\lambda_1 = 0$), while $\|\mathbf{F}\|_F^2 > 0$ implies $\lambda_M > 0$. Summing $0 \leq \lambda_m^p \leq \lambda_M^p$ over $m \in \{2, \ldots, M\}$ yields $\lambda_M^p \leq \sum_{m=2}^{M} \lambda_m^p \leq (M-1)\lambda_M^p$; inverting and multiplying by λ_M^p gives (3) and its equality conditions, and nonzero scalar rescaling cancels. □

D. Stage 4: Circuit-Breaker Panel

Instead of adding all 224 dials into one score, Stage 4 (Algorithm 1) gives each dial its own threshold, learned from the healthy calibration block. A dial *trips* when it exceeds its threshold, and an alarm fires when enough dials trip at once. Because all dials come from the same 4×32 error matrix, their trips are correlated (median effective rank 65 of 224 on the development cohort; the variance of the trip count is $7.2\times$ that of independent dials). We therefore do not assume independence; the alarm quorums are set from the trip counts actually observed on healthy calibration data:

1) *Per-dial thresholds and hijack guards:* Each dial $j \in \{1, \ldots, 224\}$ is standardized by its calibration median m_j and scaled MAD $s_j = 1.4826\ \text{MAD}(d_{j,\text{cal}}) + 10^{-4}$ (the 10^{-4} floor handles zero-MAD dials), giving $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. Its tripwire τ_j is the 98.5th calibration percentile of z_j, widened by $\gamma_j^{\text{cal}} \geq 1$ when its calibration tail grows (second-half $q_{0.98}$ above $1.35\times$ the first half) and by $\gamma_j^{\text{sc}} \geq 1$ when its scale exceeds $1.875\times$ its perspective-group median:
$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\text{cal}})\, \gamma_j^{\text{cal}}\, \gamma_j^{\text{sc}}. \tag{4}$$
2) *32 onset relays:* When a sudden fault first enters the right edge of the target window, the short prefixes ($m \in \{1, 2\}$) still cover healthy data, so few position dials trip. To catch this moment, each coordinate c has stricter 99.3rd-percentile edge tripwires ($\gamma^{1.25}$ guard) and fires a relay $r_c(t) \in \{0, 1\}$ when its full-window position dial ($m = 4$) trips together with its endpoint, fluctuation, or $m = 3$ dial.
3) *Alarm timing and quorum voting:* Under *causal attribution*, the trips from window $[t, t+S)$ are reported at its right end $\tau = t + S - 1$, the earliest moment all of its data exist; this fixes a reporting lag of $S - 1$ steps, separate from the detection delay $\Delta t = t_{\text{alm}} - t^*$ after fault onset t^*. Under *buffered attribution*, used for after-the-fact review, each trip is assigned to the steps it describes: position trips to their prefix $[t, t+S_m)$, drift and fluctuation trips to $[t, t+S)$, and endpoint trips to $[t+S_3, t+S)$. Counting active trips at each step gives $N_{224}(\tau)$, the position sub-count $N_{128}(\tau) \subset N_{224}(\tau)$, and the relay count $N_{\text{edge}}(\tau)$. For ranking metrics, the continuous score $N_{\text{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\text{edge}}(\tau)$ counts position trips and relays twice (fixed on development streams before held-out testing; Section VI-A). The binary alarm fires when any count exceeds its quorum:
$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \ \vee\ N_{128}(\tau) > K_{128} \ \vee\ N_{\text{edge}}(\tau) > K_{\text{edge}}\big), \tag{5}$$
with quorums set on calibration counts: $K_{224} = \max(1.12\, q_{0.995}, 5.0)$, $K_{128} = \max(1.12\, q_{0.995}, 3.5)$, and

$K_{\text{edge}} = \max(1.25\, q_{0.999}, 2.5)$. All quantiles use linear interpolation, the safety factor is applied before the floor, quorums stay real-valued, and both trips and alarms use strict inequalities.

E. *Compact Variant:* Laminar-JEPA-Lite

Laminar-JEPA-Lite keeps the same four stages but makes three simplifications, each supported by the ablations in Section VI. Together they cut trainable parameters by 35%–52% (141.9k at $K < 25$ and 222.3k at $K \geq 25$, vs. 295.8k), speed up windowed GPU inference by 2.43× (3.14× at $K < 25$), and need no weight editing:

1) *Smaller encoder:* A fifth dilated block blurred short-lived anomalies without making the summaries richer, so *Laminar-JEPA-Lite* uses four TCN blocks (dilations $\{1, 2, 4, 8\}$) and 64D pooling heads. The frequency-band front-end is used only on high-channel streams ($K \geq 25$), where it helped; smaller streams use a direct causal projection.
2) *Parallel three-horizon predictor:* One residual two-layer GELU MLP predicts $M = 3$ prefixes ($S_m \in \{16, 32, 64\}$) in parallel from $z_{\text{ctx}} + \mathbf{p}_m$, with a single final LayerNorm. This removes the lag of sequential prediction and the coupling between stacked normalizations, and shrinks $\boldsymbol{\Omega}$ to 96×96.
3) *128-dial panel with one quorum:* Only the 96 position dials $|\tilde{e}_{m,c}(t)|$ and the 32 drift dials $|\bar{e}_c(t)|$ are kept, each calibrated by (4). Fluctuation and endpoint dials, onset relays, and quenching are removed. Steps are ranked by the plain count $N_{128}(\tau)$, and the alarm uses one quorum, $K_{\text{Lite}} = \max\big(1.12\, q_{0.995}(N_{128,\text{cal}}),\, 3.5\big)$.

1D single-score controls: To measure what the breaker panel adds over a single score on the same backbone (Section VI), the 1D controls pool a residual vector into one number: an equal blend of its ℓ_2 norm and a softmax-weighted coordinate RMS ($\alpha = \tau_g = 0.5$). Single-horizon TS-JEPA pools $\mathbf{e}(t) \in \mathbb{R}^{32}$; 1D Multi-Horizon JEPA combines the pooled full-tensor and peak-horizon scores by root-mean-square (with optional R_{LT} modulation $\alpha_{\text{LT}} \in \{0.12, 0\}$). Both are thresholded with SPOT [8], [38], [51].

V. Experimental Protocol and Evaluation Metrics

A. Setup and Calibration Protocol

All architectures use context $C = 256$, target horizon $S = 64$ ($W = 320$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), latent dimension $D = 32$, nominal training stride 16, and evaluation stride 1. Models are trained with AdamW [52] (learning rate 10^{-3}, weight decay 10^{-4}) and zero per-dataset tuning: batch 16 for 15–20 epochs on development cohorts, and batch 16–64 for 10–12 epochs with gradient clipping (1.0) and cosine annealing on archive cohorts.

a) Disjoint Calibration Protocol: Each stream's nominal history $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Table I) into a model-fitting prefix $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = T_{\text{nom}} - \lfloor 0.35\, T_{\text{nom}} \rfloor$) and a held-out nominal suffix

Algorithm 1 Laminar-JEPA (224-Bit) and Laminar-JEPA-Lite (128-Bit)

Require: Nominal history $\mathbf{X}_{\text{nom}}$, test stream $\mathbf{X}_{\text{test}}$, tier $\in$ {Full, Lite}, attribution mode $\in$ {causal, buffered}.
Ensure: Per-timestep binary alarms $\hat{y}(\tau) \in \{0, 1\}$ and continuous score $N_{\text{tot}}(\tau)$.

Phase 1: Disjoint Training, Whitening & Adaptation ($\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{sel}}, \mathbf{X}_q$)
Split $\mathbf{X}_{\text{nom}}$ chronologically into S-isolated ($\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{sel}}, \mathbf{X}_q, \mathbf{X}_{\text{cal}}, \mathbf{X}_{\text{hold}}$) and standardize channels using $\mathbf{X}_{\text{fit}}$ alone.
Set nested prefixes $S_{1:4} = \{16, 32, 48, 64\}$ (Full) or $S_{1:3} = \{16, 32, 64\}$ with $K \geq 25$ rFFT gate (Lite).
for stride-16 training batches in $\mathbf{X}_{\text{fit}}$ **do**
    Encode $z_{\text{ctx}} = E_\theta(\mathbf{x}_{\text{ctx}})$ and targets $z_{\text{tgt}}^{(m)} = E_\phi(\mathbf{x}_{\text{tgt}}^{(m)})$ with barycenter shift $-\bar{\boldsymbol{\mu}}_{\text{ctx}}$.
    Roll out sequential $\{P_{\text{sub}}, P_\theta\}$ (Full) or parallel single-LayerNorm predictor (Lite); update θ via (1) and $\phi \leftarrow m\phi + (1-m)\theta$.
end for
Select checkpoint on $\mathbf{X}_{\text{sel}}$ (or fixed epochs) and fit $(\bar{\boldsymbol{\mu}}_{\text{res}}, \boldsymbol{\Omega} \in \mathcal{S}_+^{MD})$ on $\mathbf{X}_{\text{fit}}$.
if tier = Full and $q_{\text{wsp}} = 1$ **then**
    Rank drifting coordinates on $(\mathbf{X}_{\text{fit}}, \mathbf{X}_q)$; run Steps Q1–Q4 for $k \in \{4, 2\}$ and keep edit iff $\mathbf{X}_q$ split and 0.92-quorum gates pass.
end if
Fit and freeze stabilization scales and blending weights $w_{m,c}$ (2) on $\mathbf{X}_q$.

Phase 2: Untouched Circuit-Breaker Calibration on $\mathbf{X}_{\text{cal}}$
Compute frozen stabilized residuals $\tilde{\mathbf{E}}_{\text{cal}}$ on untouched $\mathbf{X}_{\text{cal}}$ via (2).
if tier = Full **then**
    Expand $\tilde{\mathbf{E}}_{\text{cal}}$ into $7D = 224$ dials (position, drift, fluctuation, endpoint) and 32 right-edge onset relays.
    Fit MAD scales (m_j, s_j), hijack guards γ_j, tripwires τ_j (4), and quorums $(K_{224}, K_{128}, K_{\text{edge}})$ (5).
else ▷ Preferred Deployment Tier: Laminar-JEPA-Lite
    Extract $4D = 128$ position-plus-drift dials $(|\tilde{e}_{m,c}|, |\bar{e}_c|)$, fit τ_j (4), and set single quorum $K_{\text{Lite}} = \max(1.12\, q_{0.995}, 3.5)$.
end if

Phase 3: Online Inference on $\mathbf{X}_{\text{hold}}$ or $\mathbf{X}_{\text{test}}$
for each stride-1 window $[t - C,\, t + S)$ **do**
    Compute $\tilde{\mathbf{E}}(t) \to \mathbf{d}(t)$ and binary trips $b_j(t) = \mathbb{I}(z_j(t) > \tau_j)$ (plus relays $r_c(t)$ if Full).
    Attribute active counts to right edge $\tau = t + S - 1$ (causal) or prefix spans $[t, t + S_m)$ (buffered).
    Emit $\hat{y}(\tau) \in \{0, 1\}$ when active counts exceed $(K_{224}, K_{128}, K_{\text{edge}})$ (5) or K_{Lite}.
end for

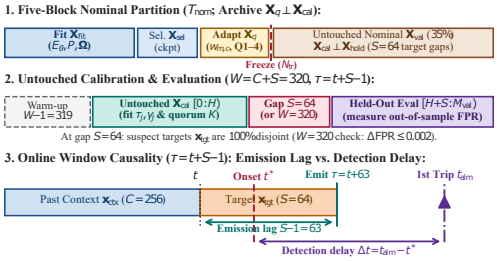


Fig. 2. Disjoint nominal evaluation protocol (C=256, S=64, W=320): (Row 1) five-block archive partition into S-isolated weight-fitting ($\mathbf{X}_{\text{fit}}$), checkpoint-selection ($\mathbf{X}_{\text{sel}}$), and nominal adaptation ($\mathbf{X}_q$) prefixes versus $\mathbf{X}_{\text{val}}$, (Row 2) untouched threshold calibration ($\mathbf{X}_{\text{cal}}$) and held-out nominal evaluation ($\mathbf{X}_{\text{hold}}$), and (Row 3) structural emission lag ($S-1$) versus empirical detection delay (Δt).

$\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$: (1) *Stage A (Disjoint weight fitting, checkpoint selection, and adaptation):* Reusing calibration windows for early stopping, blending (2), or quenching can make nominal thresholds optimistic. Archive benchmarks (Table VI) therefore use a disjoint *five-block protocol* ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$) inside $\mathbf{X}_{\text{tr}}$: $\mathbf{X}_{\text{fit}}$ ($\approx$ 43% of T_{nom}) fits standardization, network weights, $\bar{\boldsymbol{\mu}}_{\text{ctx}}$, and $\boldsymbol{\Omega} \in \mathcal{S}_+^{MD}$; $\mathbf{X}_{\text{sel}}$ ($N_{\text{sel}} \geq 96$ windows) selects the checkpoint; and $\mathbf{X}_q$ ($N_q \in [160, 2{,}048]$ stride-1 windows isolated from $\mathbf{X}_{\text{sel}}$ and $\mathbf{X}_{\text{val}}$ by $S = 64$ steps) fits and freezes (2) and optional

TABLE I
NOMINAL PARTITIONING AND CALIBRATION PROTOCOL BY COHORT

Cohort (Tables)	Weights (%)	Ckpt. Sel.	Adapt. / Quench	Thresholds
Dev. 19 Cache (IIa,b, III)	$\mathbf{X}_{\text{tr}}$ (65)	Fixed ep.	None (unblended)	$\mathbf{X}_{\text{cal}}$
Dev. 33 Suite (IIc, IV, V)	$\mathbf{X}_{\text{tr}}$ (65)	Fixed ep.	Shared $\mathbf{X}_{\text{cal}}$	$\mathbf{X}_{\text{cal}}$
Dev. 19 4-Block (Sec. V-B)	$\mathbf{X}_{\text{fit}}$ (50)	Fixed ep.	Disjoint $\mathbf{X}_q$	$\mathbf{X}_{\text{cal}}$
Archive 118/141 (VI)	$\mathbf{X}_{\text{fit}}$ (43)	$\mathbf{X}_{\text{sel}}$	Disjoint $\mathbf{X}_q$	$\mathbf{X}_{\text{cal}}$
Held-Out / Test ($\mathbf{X}_{\text{hold}}$, $\mathbf{X}_{\text{test}}$)	Frozen	Frozen	Frozen	Frozen

quenching (Steps Q1–Q4) before $\mathbf{X}_{\text{cal}}$ is touched. Fixed-epoch development runs (Table I) omit $\mathbf{X}_{\text{sel}}$, and the 19-stream cache skips (2) and quenching. A four-block check ($\mathbf{X}_{\text{fit}} \to \mathbf{X}_q \to \mathbf{X}_{\text{cal}} \to \mathbf{X}_{\text{hold}}$) on the 19 development streams gives causal FPR_{hold} of 0.96% (*Lite*) and 1.05% (*Laminar-JEPA*), matching Table IV, though the smaller 50% fitting block lowers causal F1 ($0.133 \to 0.071$, $0.144 \to 0.076$). On short histories, *two-fold cross-fitting* swaps the halves of $\mathbf{X}_{\text{val}}$ and requires $\hat{y}^{(1)} \wedge \hat{y}^{(2)}$ (Section VII-E). (2) *Stage B (Untouched calibration and boundary isolation on $\mathbf{X}_{val}$):* Stride-1 windows of length $W = 320$ in $\mathbf{X}_{\text{val}}$ give $M_{\text{val}} = N_{\text{val}} - W + 1$ indices. Splitting at $H = \lfloor M_{\text{val}}/2 \rfloor$ defines $\mathbf{X}_{\text{cal}} = [0 : H)$ and $\mathbf{X}_{\text{hold}} = [H + S : M_{\text{val}})$, separated by an $S = 64$-step gap $[H : H + S)$ so target observations never overlap (a $G = 320$ gap shifts causal FPR by < 0.25 points). On $\mathbf{X}_{\text{cal}}$ alone, the frozen model fits all dial scales, guards, tripwires (4), and quorums (5) (or τ_{SPOT} for 1D baselines), frozen on $\mathbf{X}_{\text{hold}}$ and $\mathbf{X}_{\text{test}}$. We separate raw count $N_{\text{cal}} = H$ from autocorrelation-adjusted *effective sample size* $N_{\text{eff},s} = N_{\text{cal}}/\tau_{\text{int},s}$ via Sokal estimators (τ_{int} up to 206; Section VII-E).

b) Statistical Inference: We report a two-level inference hierarchy after averaging seeds within each stream (seed standard deviation (SD)): (1) *Dataset-clustered and stream-level inference: Dataset-clustered inference* (95% cluster-bootstrap CIs and exact 2^D dataset sign-flip permutation tests) is emphasized alongside stream-level paired Wilcoxon tests and 95% stream-bootstrap CIs (2,000 resamples). For the four focal causal contrasts on the 118 untouched streams (*Laminar-JEPA-Lite* vs. Multi-Horizon JEPA (1D) and TS-JEPA, on Point-F1 and ER), Benjamini–Hochberg (BH) [41] adjusted p-values are 0.014 and 3.6×10^{-4} (stream-level Point-F1) and 0.0078 and 0.012 (dataset sign-flip ER, positive on $9/9$ and $8/9$ datasets); other quoted p-values are raw and descriptive, and non-Exathlon subsets ($n = 101$, 8 datasets) verify sensitivity to archive composition. (2) *12-fold LODO audit:* Across all 12 LODO folds of the 141-stream archive, *Laminar-JEPA-Lite* keeps a positive gain on causal ER ($+0.099$ to $+0.141$ vs. Multi-Horizon 1D), buffered Point-F1, and causal Point-F1 vs. TS-JEPA ($\max p = 0.0054$); its causal Point-F1 gain over Multi-Horizon 1D holds in $9/12$ folds.

B. Cohorts and Baselines

Laminar-JEPA (w/o Quench) omits Steps Q1–Q4. Causal and buffered modes share identical splits per cohort (except a single-horizon 130-stream buffered screen in Section VII-A): (1) *Development cohorts (design-stage, Table I):* Because these streams were inspected during design and the 33-stream suite shares $\mathbf{X}_{\text{cal}}$ for stabilization and thresholds, they compare design choices rather than out-of-sample calibration. The 19-stream cache (6 datasets) tests the dial ladder, panel components, and conformal/SPC banks on shared unquenched residuals (Table II(a,b), Table III); the 33-stream benchmark (7 datasets: 19 non-GHL in Table II(c) plus 14 GHL in Table IV) compares all baselines. Tripwire and quench constants were fixed on 23 tuning streams (19 non-GHL plus `SMAP T-2`, `MSL P-10`, `Daphnet S09R01E2`, `OPPORTUNITY S1-ADL3`). (2) *Untouched held-out cohorts (primary evidence):* Removing those 23 tuning streams from the 141-stream archive (12 datasets, five-block protocol) leaves 118 untouched streams across 9 datasets (101 non-Exathlon across 8 datasets; Table VI), plus a single-seed Q1–Q4 isolation audit (Section VI-A).

Three baseline tiers: (i) *Matched-backbone controls* (TS-JEPA, Multi-Horizon JEPA (1D)) fix the representation and test one SPOT score against the panel. (ii) *System-level aggregators* fix the whitened residuals and replace the panel with maximum z, top-k voting, ℓ_2^2 conformal, Fisher/Stouffer p-pooling, multiplicity-controlled conformal banks, or SPC charts, calibrated to the panel's $\mathbf{X}_{\text{cal}}$ alarm rate unless noted (Tables III, VI). (iii) *External baselines* (TCN Forecaster/Autoencoder, PCA Forecaster, Isolation Forest, TimesNet [5], TranAD [6], Anomaly Transformer [7], NCAD [10], DCdetector [17]) share splits, training-only scaling, gaps, and SPOT ($u = p_{98}$, $q = 10^{-3}$), using 206k–649k parameters (panels add none). Runs used PyTorch 2.4 / CUDA 12.4 on an RTX 5090 (0.15 ms/window *Lite*, 0.38 ms *Laminar-JEPA*; 188 MB peak); code: https://github.com/quaid281/NCAD.

C. Metrics

Every alarm $\hat{y}(\tau) \in \{0, 1\}$ and continuous score $N_{\text{tot}}(\tau)$ is scored step by step against $y(\tau) \in \{0, 1\}$ without point adjustment [11]. *Causal* numbers show what an operator sees live at step τ; *buffered* numbers reflect retrospective review after the target window finishes and are not live performance. Headline values average seeds per stream before averaging across streams (or unweighted across datasets in dataset-macro Buffered Point-F1, DS-F1):

1) *False-alarm rates (FPR_{hold}, FPR_{test}):* Fraction of healthy steps flagged on the untouched pre-test check block $\mathbf{X}_{\text{hold}}$ ($[H + S : M_{\text{val}})$) and on healthy test steps ($y(\tau) = 0$). Because a few drifting streams skew the mean, we also report stream medians and 90th percentiles (p90).
2) *Step-level accuracy (Point-F1, precision-recall (PR) AUC):* Point-F1 (Pt-F1) is the harmonic mean of step precision (fraction of alarm steps that are truly faulty) and step recall (fraction of faulty steps flagged). Without point adjustment, Pt-F1 is naturally modest (0.05–0.25, not 0.90): residuals peak near regime changes rather than across every step of a long fault, and when faults occupy a small fraction π of the stream, even a 1% false-alarm rate adds many false-positive steps (alarming at every step scores $\text{F1}_{\text{all}} = 2\pi/(1 + \pi)$). PR-AUC (Buf. PR when buffered) scores ranking quality across all thresholds without inflation from healthy steps.
3) *Episode catch rate and speed (ER, Delay Δt):* Event recall (ER) is the fraction of contiguous faulty episodes ($y = 1$ runs) with at least one alarm; it must be read with FPR because random alarms at rate p still hit long episodes at rate $\text{ER}_{\text{rand}}(p)$. Detection delay $\Delta t = t_{\text{alm}} - t^*$ counts

steps from fault onset t^* to the first in-segment alarm over *detected* episodes (distinct from the $S-1$ causal lag), so catching an extra late-developing fault can raise mean Δt.

4) *Budget controls and score diagnostics: Auto.* uses the deployed healthy-only threshold (quorums or SPOT), $q^* = 0.05$ sets thresholds at the 95th $\mathbf{X}_{\text{cal}}$ percentile (q_{1-q^*}), and diagnostic controls $f^* \in \{0.01, 0.05\}$ use test labels post-hoc to match a 1% or 5% FPR_{test}. On nominal scores, excess kurtosis γ_2 ($\gamma_2 = 0$ for Gaussian tails) measures $\mathbf{X}_{\text{cal}}$ spike heaviness, and the two-sample Kolmogorov–Smirnov (KS) distance $D_{\text{KS}} \in [0,1]$ ($\mathbf{X}_{\text{cal}}$ vs. $\mathbf{X}_{\text{hold}}$) measures healthy drift.

VI. Ablation Studies and Analysis

A. Ablation 1: Panel Components and Quenching

Table II(a),(b) and Fig. 3(a) evaluate perspective expansion and panel components on the 19-stream residual cache (not paired to the separately trained run in Panel (c)):

1) *Scientific role of the 224-dial reference vs. compact panels:* More dials do not make a better operational detector; we keep 224-dial *Laminar-JEPA* as a diagnostic reference because scoring all four views on one residual tensor isolates which views aid live causal versus buffered attribution. Against 1D scoring on this cache, 224 dials raise causal Point-F1 from 0.013 to 0.071 ($p = 0.0023$; buffered $+0.033$, $p = 0.53$), whereas pruning fluctuation, endpoint, and relay dials to 160 position-plus-drift dials raises causal Point-F1 to 0.098, ER to 59.1%, and cuts delay ($90.1 \rightarrow 74.8$ steps) and FPR_{hold} ($0.76\% \rightarrow 0.55\%$) while lowering buffered Point-F1 ($0.224 \rightarrow 0.198$). Retraining on the 118 untouched archive streams (Table VI(b)) tests 224-dial *Laminar-JEPA*, same-run 160-dial *Laminar-JEPA*, and 3-horizon *Laminar-JEPA-Lite* (128 dials) on identical splits: pruning $224 \rightarrow 160$ dials cuts buffered FPR_{hold} by 6.2 points ($19.2\% \rightarrow 13.0\%$, $p = 3.7\times10^{-15}$) at near-identical causal Point-F1 (0.073 vs. 0.074), while *Laminar-JEPA-Lite* matches or exceeds both (0.078 causal, 0.154 buffered Point-F1) at lower cost.

2) *Relays, PR-AUC score weighting, quorum rules, and hijack guards:* Removing the 32 onset relays lowers causal Point-F1 ($0.071 \rightarrow 0.068$), and replacing the three-counter quorum (5) with one 224-dial quorum lowers causal Point-F1 to 0.063 ($\text{FPR}_{\text{hold}} = 0.0055$; Table III). Replacing the double-position continuous score $N_{\text{tot}} = N_{224} + N_{128} + 2N_{\text{edge}}$ with unweighted $N_{224}+N_{\text{edge}}$ or N_{224} leaves binary alarms unchanged and changes PR-AUC by at most 0.004 on this cache and 0.0016 on the 118 untouched streams (*Laminar-JEPA-Lite* uses unweighted N_{128}). Removing hijack guards ($\gamma_j = 1$) raises causal Point-F1 to 0.073 (70.1-step delay) on this cache when nominal drift is mild, so the guards are a safeguard against drift rather than a source of accuracy. No panel component removes the $S-1$ reporting lag.

3) *Quenching step isolation and LayerNorm coupling:* We tested each quench step in isolation on single-seed *Laminar-JEPA* checkpoints across all 118 untouched streams (unquenched: causal $\text{FPR}_{\text{hold}} = 6.76\%$, F1 $= 0.0901$). Damping $k = 4$ predictor rows (Step Q1) shifts unselected coordinates by a median 11.1% of nominal RMS through 32D LayerNorm coupling and does not lower FPR_{hold} (7.06%). Recentering and decoupling (Steps Q2–Q3) lower it to 6.71% but cut causal F1 to 0.0826. The split gate (Step Q4, accepted on $39/118$ streams) reaches 6.62% with F1 unchanged (0.0904). Because ungated edits hurt $k = 2$ causal F1 on $11/19$ development streams ($0.141 \rightarrow 0.049$) and $k \in \{8, 12\}$ cuts PR-AUC ($0.292 \rightarrow 0.223$), *Laminar-JEPA-Lite* omits weight editing altogether.

Marginal, multiplicity-controlled, and system-matched detector banks: Simple per-coordinate alarms either fire constantly without a quorum, almost never fire under multiplicity correction on short calibration sets, or lose their calibration out of sample. We compare three calibration regimes on the same 19-stream whitened residuals without test-label tuning (Table III): (1) *Per-coordinate marginal union:* Thresholding 128 coordinates individually at $q_{0.985}$ or even at the panel's 0.34% rate without a quorum compounds false alarms across coordinates ($\text{FPR}_{\text{hold}} = 66.5\%$ and 40.6%). (2) *Family-wise and false discovery rate (FDR) multiplicity control:* Applying Šidák (0.5%) across 128 coordinates or BH FDR ($q = 0.005$) across 224 dials hits the discrete conformal p-value floor $(N_{\text{cal}} + 1)^{-1} > 3.9 \times 10^{-5}$, paralyzing causal recall (F1 ≤ 0.001). (3) *System-level matched nominal alarm rate:* Calibrating the aggregated statistic directly to each stream's nominal panel rate (0.34%) shows that discrete conformal $\min_j p_j$ still jumps to $\text{FPR}_{\text{hold}} = 21.1\%$ out of sample from boundary ties, and recalibrated Mahalanobis conformal, Hotelling T^2, MEWMA, and CUSUM inflate to 27.9%–58.6%. Among stable rules on this cache, the 160-dial panel (F1 $= 0.098$, $\text{FPR}_{\text{hold}} = 0.55\%$) beats 128-coordinate top-k (0.068), $\max_j z_j$ (0.056), and ℓ_2^2 conformal (0.072), whereas Fisher pooling reaches higher Point-F1 (0.109; Stouffer 0.095) at nearly double the held-out FPR (1.05% vs. 0.55%). On the 118 untouched archive streams using *Laminar-JEPA-Lite*'s frozen residuals (Table VI(b)), *Laminar-JEPA-Lite* (0.078 F1, 65.0% ER, 7.1% FPR_{hold}) beats Fisher (0.077 F1, 61.1% ER at 8.0% FPR_{hold}; $+3.9$ ER points on $9/9$ datasets, sign-flip $p = 0.0078$), Stouffer (0.057 F1, 55.8% ER), raw top-k (0.069 F1, 58.1% ER; $+6.9$ ER points on $9/9$ datasets), $\max_j z_j$ (0.063 F1, 58.8% ER), ℓ_2^2 conformal (0.071 F1, 55.6% ER), and BH-FDR conformal (0.024 F1, 14.6% ER).

B. Ablation 2: Single-Channel Faults and Complementary Channel-Level Anomaly Branch

A frozen or flatlined sensor barely moves the latent summary: terminal LayerNorm spreads the small change across all 32 coordinates, so on `SMAP P-3` a flatline trips 12.5 dials versus 62.1 for a variance-matched harmonic flip, and a single faulty sensor is further diluted by the $K-1$ healthy channels. Extending Fig. 3(b) (n=6), we evaluate a causal channel branch across all 19 multivariate development streams ($K \in [5, 55]$) without per-stream or test-label tuning. Per channel, three 64-step rolling statistics—log-variance drop $-\ln(\text{Var}_{64}(x_c) + 10^{-4})$, training-fit first-order autoregressive (AR(1)) residual RMS, and mean shift $|\bar{x}_{64,c} - \mu_c|/\sigma_c$—are thresholded at $1.25\, q_{0.999}$ on nominal validation windows

TABLE II
CIRCUIT-BREAKER DIAL EXPANSION, COMPONENT ABLATIONS, AND MATCHED-BACKBONE STUDY

(a) Perspective Expansion Ladder (1 → 224 Dials, n=19)

Perspective Expansion Stage	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
1D Scalar Collapse (1 Dial)	.013	.191	**.315**	.413	.459	**.550**	166.2	143.1
32-Bit Full Horizon (m=4)	.050	.135	.255	.381	.500	.243	123.4	65.5
128-Bit Sub-Horizons (m=1..4)	.063	.216	.294	**.449**	.556	.462	110.6	**38.8**
160-Bit (+32 Horizon Drift)	**.098**	.198	.304	.429	**.591**	**.550**	**74.8**	108.3
192-Bit (+32 Horizon Fluc.)	.067	.192	.299	.443	.573	.453	79.1	101.6
224-Bit (+32 Endpoint/Relays)	.071*	**.224**	.299	.448	.564	.506	90.1	83.7

* Versus 1D, causal F1 difference $+0.057$ (stream-bootstrap 95% CI $[0.016, 0.120]$; Wilcoxon $p = 0.0023$, 12/2 wins/losses). Buffered difference $+0.033$ (CI $[-0.017, 0.093]$, $p = 0.53$). The scalar spectral-modulation ablation in the separately trained study remains null.

(b) Circuit-Breaker Component & Quorum Ablation (n=19)

Component / Quorum Variant	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
224-Bit Breaker (Full Panel)	.071	.224	.299	.448	**.565**	**.506**	90.1	83.7
224-Bit Strict (High Quorum K_{strict})	.047	.168	.299	.448	.494	.439	207.6	**54.3**
w/o 32 Right-Edge Relays	.068	.225	.299	**.452**	**.565**	**.506**	90.7	83.7
w/o Prefix Routing (Unif. Smear)	.071	.222	.299	.440	**.565**	.497	90.1	115.4
w/o Hijack Guards (γ_j=1)	**.073**	**.226**	**.299**	.448	**.565**	**.506**	**70.1**	83.3

Panels (a,b): 19 streams × 3 seeds on the same whitened-residual cache, without the stabilized blend in (2). Panel (c) uses a separate training run; cross-panel comparisons are not paired.

(c) Matched-Backbone Architectural Comparison (n=19 Development Streams)

Architecture Arm	Tail / Drift γ_2	Tail / Drift D_{KS}	Held-Out FPR Caus.	Held-Out FPR Buf.	Point-F1 Caus.	Point-F1 Buf.	PR-AUC Caus.	PR-AUC Buf.	Causal Event Recall Auto.	Causal Event Recall q^*=0.05	Causal Event Recall f^*=0.01	Causal Event Recall f^*=0.05
Laminar-JEPA (224-Bit)	0.92	0.190	0.013	0.128	0.144	**0.244**	**0.311**	**0.407**	0.673	0.795	0.612	0.757
Laminar-JEPA (w/o Quench)	0.88	0.190	0.016	0.128	0.142	0.235	0.304	0.381	**0.695**	0.815	**0.631**	0.765
Laminar-JEPA-Lite (128-Bit)	2.33	0.182	0.010	0.061	0.133	0.242	0.301	0.397	0.650	0.811	0.560	**0.772**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Multi-Horizon JEPA (1D)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Multi-Horizon JEPA (α_{LT}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Frozen Target	1.16	**0.167**	0.009	0.098	0.146	0.235	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

Panel (c) (5 seeds): γ_2 = excess kurtosis on $\mathbf{X}_{\text{cal}}$; D_{KS} = KS drift ($\mathbf{X}_{\text{cal}}$ vs. $\mathbf{X}_{\text{hold}}$); Auto. = deployed healthy-calibrated threshold; q^*=0.05 = 95th-percentile $\mathbf{X}_{\text{cal}}$ threshold; f^* = post-hoc diagnostic FPR_{test} control.

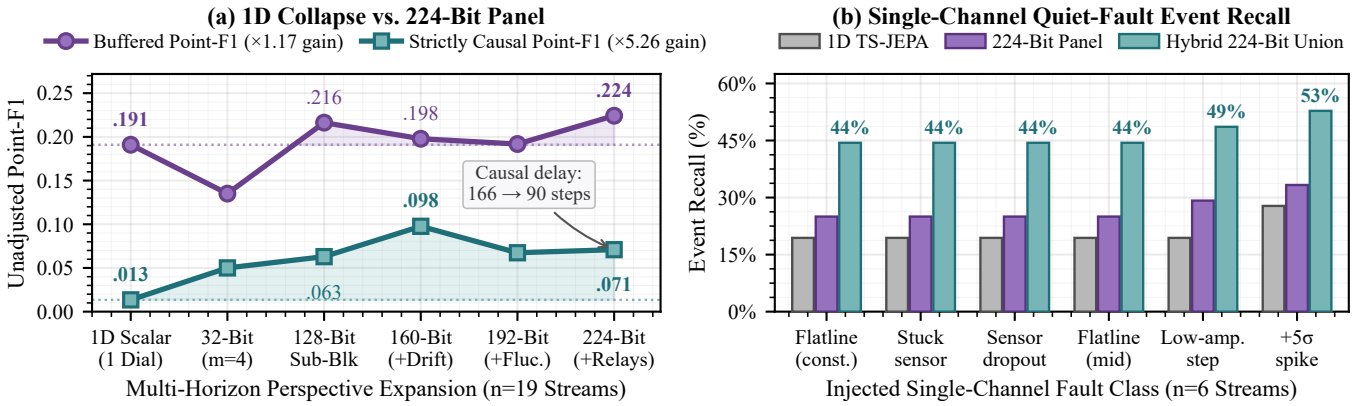


Fig. 3. Development studies. **(a)** Causal and buffered Point-F1 across the six-step expansion on cached whitened residuals from 19 streams (three seeds; no stabilization blend). **(b)** Event recall for injected single-channel faults in a separate six-stream study, comparing 1D TS-JEPA, 224-Bit Breaker, and their channel-level union.

(floor 6.0 MAD, 1.25× half-split drift multiplier) with a 24-of-40-step persistence gate. On the highest-variance active channel $c^\star = \arg\max_{c:\sigma_c^{\text{tr}}>10^{-8}} \sigma_c^{\text{tr}}$, we place three nonoverlapping 200-step faults per stream inside clean test segments (57 events per class). For $s \le t < s+200$ using training $(\mu_c, \sigma_c, q_{.995,c})$:

$$x'_{t,c^\star} = \begin{cases} \mu_{c^\star} & \text{variance collapse,} \\ \mu_{c^\star} + 1.35(-1)^{t-s}\sigma_{c^\star} & \text{prediction shock,} \\ x_{s-1,c^\star} & \text{sensor freezing,} \\ \max(q_{.995,c^\star}, \mu_{c^\star} + 2.2\sigma_{c^\star}) & \text{saturation,} \\ x_{t,c^\star} + 2.5\sigma_{c^\star}(t-s)/199 & \text{gradual degradation,} \\ x_{t,c^\star} + 1.5\sigma_{c^\star} & \text{isolated bias.} \end{cases} \tag{6}$$

Outside $[s, s+200)$, all channels remain unchanged. The variance-collapse union lifts injected-event recall for variance collapse, freezing, and saturation from 63.2%–82.5% to 93.0%–94.7%, and the full channel branch lifts recall on shocks, gradual degradation, and bias from 59.6%–75.4% to 80.7%–96.5%. However, on the *unmodified* test streams, causal FPR_{test} rises from 1.5% (standalone panel, F1 = 0.071) to 6.8% (variance-collapse union, F1 = 0.070) and 19.4% (full branch, F1 = 0.129). Synthetic single-channel injections do not establish detection on natural sensor failures, and the full union's 19.4% false-alarm rate makes it unsuitable as a default detector.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

Retraining the multi-horizon JEPA backbone with and without 4-shell rFFT splitting on five streams (`SMAP P-3`, `MSL M-1`, `SMD machine-1-2`, `Daphnet S01R01E1`, `GECCO`; three seeds) shows a channel-count threshold: shell splitting improves high-channel spacecraft detection (K=55 `MSL M-1` buffered F1 0.085 → 0.410; K=25 `SMAP P-3` causal F1

TABLE III
COORDINATE DETECTOR-BANK REGIMES ON CACHED RESIDUALS

Detector & Calibration Regime	Pt-F1 ↑	ER ↑	$\mathrm{FPR}_{\mathrm{hold}}$ ↓	$\mathrm{FPR}_{\mathrm{test}}$ ↓
Multi-horizon circuit-breaker panels (nominal quorum)				
160-dial pos.+drift (single quorum)	**0.098**	0.591	0.0055	0.0087
224-dial panel (deployed)	0.071	0.564	0.0076	0.0150
224-dial single quorum (no relay)	0.063	0.564	0.0055	0.0088
1. Per-coordinate marginal union (no quorum, 128 coords)				
Per-coord. $q_{.985}$ union	0.291	0.913	0.6651	0.7353
Per-coord. .0034 marginal union	0.303	0.874	0.4057	0.4765
2. Family-wise / FDR multiplicity control (α, q=.005)				
Per-coord. conformal (Šidák)	0.000	0.000	**0.0000**	**0.0000**
224-dial conformal BH FDR	0.000	0.026	**0.0000**	**0.0000**
3. System-level matched nominal rate (mean .0034 on $\mathbf{X}_{cal}$)				
Raw 128-coord. top-k vote	0.068	0.564	0.0052	0.0096
Continuous 128-coord. $\max_j z_j$	0.056	0.658	0.0078	0.0147
Discrete conformal $\min_j p_j$	0.265	0.843	0.2111	0.2831
Multivar. conf. ($\|\mathbf{e}_{\mathrm{white}}\|_2^2$)	0.072	0.608	0.0062	0.0095
Fisher p-pool (224 dials)	0.109	0.643	0.0105	0.0183
Maximum 224-dial z	0.083	0.652	0.0093	0.0157
Hotelling T^2 (recalibrated)	0.283	0.816	0.2900	0.4831

19 streams × 3 seeds, causal means. System-matched rules match each stream's nominal panel alert rate after aggregation (Stouffer .095 F1, .0087 $\mathrm{FPR}_{\mathrm{hold}}$; 224-dial top-$k$ .070, .0077; Mahalanobis conf. .2789, CUSUM .3457, MEWMA .5862 $\mathrm{FPR}_{\mathrm{hold}}$).

0.212 → 0.321), but lowers F1 on both K=9 streams (`Daphnet`, `GECCO`), setting the $K \geq 25$ spectral gate in *Laminar-JEPA-Lite*.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing all 16 matched-budget arms in Table II(c) on the development streams separates five contributing factors under these protocols (not universal causes): (1) raw-observation forecasting and autoencoding share heavy tails ($\gamma_2 \approx 6.2$), large drift ($D_{\mathrm{KS}} \approx 0.45$), and $\approx 25\%$ causal FPR; (2) a D=32 LayerNorm bottleneck in observation space cuts kurtosis ($\gamma_2 = 0.91$) and halves FPR, yet raw targets keep drift high ($D_{\mathrm{KS}} = 0.376$); (3) learned D=32 latent targets and whitening halve drift ($D_{\mathrm{KS}} = 0.214$ in TS-JEPA and 0.179–0.190 across multi-horizon JEPA tiers, $p \leq 0.001$); (4) linear PCA and Isolation Forest keep low causal FPR (0.021 and 0.039) but reach only 0.017 and 0.041 causal Point-F1; and (5) frozen targets and ℓ_2 scoring retain $< 2\%$ causal FPR, while latent context autoencoding lowers causal Point-F1 (0.112 vs. 0.161 for TS-JEPA). The panels' gains do not come from lighter tails alone: *Laminar-JEPA-Lite* keeps roughly the same kurtosis as Multi-Horizon JEPA (1D) ($\gamma_2 = 2.33$ vs. 2.30).

VII. RESULTS AND DISCUSSION

A. Design-Stage Benchmark and Scope Boundaries

Following Section V-B, Table IV reports exploratory design-stage results on the 33-stream development suite (where stabilization and thresholds share $\mathbf{X}_{\mathrm{cal}}$). Table VI provides our primary generalization and calibration evidence under the disjoint five-block protocol across 118 untouched streams and all 141 archive streams. On a separate 130-stream buffered screen, TS-JEPA reaches stream-mean Point-F1 0.186 versus TranAD 0.152, TimesNet 0.151, DCdetector 0.097, NCAD 0.088, and Anomaly Transformer 0.082. Spherical normalization alone gives no distributional guarantee; lower false alarms with latent scoring hold specifically under these protocols.

Design-stage configuration guidance: Both tiers reach nearly identical unweighted buffered Point-F1 (0.244 vs. 0.242) and causal ER (67.3% vs. 65.0%) on the 33-stream suite. *Laminar-JEPA-Lite*, however, halves buffered $\mathrm{FPR}_{\mathrm{hold}}$ (6.1% vs. 12.8%) and leads 7-dataset macro Point-F1 (0.171 vs. 0.133). We deploy *Laminar-JEPA-Lite* operationally and keep *Laminar-JEPA* for ablation.

Slow-ramp boundary: A $C = 256$-step encoder cannot see drift slower than its window. Context and target summaries move together, yielding 0.000 Point-F1 on all 14 GHL slow-ramp streams. Injecting linear ramps of amplitude $a \in \{0.5, 1, 2, 4\}\sigma$ and length $L \in \{64, 256, 1024, 4096\}$ (15 events per cell) maps the cutoff: steep multi-channel ramps ($L = 64$, $a \geq 1\sigma$) reach 86.7%–88.7% ER at 46–54-step median delay (5.7–7.7 points above background chance), $L = C$ falls to 0.5–2.5 points above chance, and for $L \geq 1024$ ($4C$) chance-corrected detection vanishes ($[-4.8, +0.8]$ points) across every JEPA variant.

B. Real-Time Streaming vs. Retrospective Buffered Attribution

Live streaming is much harder than retrospective review. Comparing Panels (a) and (b) of Table IV shows why 1D latent scores degrade under causal reporting: buffered attribution averages residuals across every overlapping window covering a timestamp, whereas causal streaming scores a single right-edge step with no averaging, allowing one or two drifting coordinates to inflate the shared SPOT threshold (Section I). On these development streams, a rate-matched Bernoulli alerter $\mathrm{ER}_{\mathrm{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$ hits 28.1% of events at $\leq 1.3\%$ causal FPR, versus 60.0% at TimesNet's 24.3%.

C. Exploratory Horizon Sweep Across $S \in \{1, 4, 16, 32, 64\}$

Shorter horizons trade buffered smoothing for lower live lag. Because a target window of length S reports causally at $\tau = t + S - 1$ (63-step lag at $S = 64$), we retrain on five exploratory streams across $S \in \{1, 4, 16, 32, 64\}$ (Fig. 4(a); at $S = 1$ all prefixes coincide, and at $S = 4$ the shortest prefix spans two steps), where both panels exceed 1D Multi-Horizon SPOT at every S with test FPR $\leq 3.6\%$. At $S = 1$ and $S = 4$, *Laminar-JEPA* reaches 2.6× and 4.3× scalar causal Point-F1 ($0.037, 0.050$ vs. $0.014, 0.012$; 77%–84% ER) and *Laminar-JEPA-Lite* reaches 2.1× and 3.1× ($0.030, 0.036$; 64%–82% ER), cutting delay at $S = 4$ to 36.7–44.1 steps, whereas buffered Point-F1 peaks at 0.290 ($S = 32$) and 0.255 ($S = 64$) versus 0.182 for 1D SPOT. Five selected streams cannot set a general rule.

D. Design-Stage Operating Budgets and Fault Taxonomies

Fig. 4(b) and Table V examine the 19 development streams across budgets and fault types: (1) *Diagnostic budgets (Fig. 4(b), Table II(c)):* At a post-hoc matched $f^* = 1\%$ test FPR, *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% (63.1% unquenched) and 56.0% of events versus 39.5% for TS-JEPA, 21.7% for TimesNet, and 14.9% for TranAD. (2) *Contextual vs. point faults (Table V):* Out-of-range spikes ($> 5\sigma$) are caught by reconstructors and amplitude rules alike, whereas in-range contextual faults alter cross-channel phase or velocity within normal bounds; on NASA contextual

TABLE IV
PRIMARY 33-STREAM DEVELOPMENT BENCHMARK ACROSS ATTRIBUTION REGIMES

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR_{hold} ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR_{hold} ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR_{hold} ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-horizon coordinate circuit-breaker architectures												
Laminar-JEPA (224-Bit)	0.128	**0.244**	**0.407**	**0.669**	432.9	0.013	0.144	**0.673**	249.9	0.004	0.000	0.133
Laminar-JEPA-Lite (128-Bit)	0.061	0.242	0.397	0.636	59.3	0.010	0.133	0.650	73.8	0.005	0.000	**0.171**
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Multi-Horizon JEPA (1D)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Multi-Horizon JEPA (α_{LT}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	$F1_{all}$	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.214	0.179	0.145	**0.229**	0.144	**.009/.061**	.012/.091	.030/.091	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.003/.004**	.005/.005	.004/.004	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	0.078	**0.191**	0.000	0.000	0.000	.009/.104	.008/.020	.004/.017	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.284**	0.210	0.161	0.158	0.151	.013/.151	**.004/.049**	.054/.095	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.357	0.375	**0.379**	0.289	0.167	.021/.223	.016/**.066**	**.014**/.135	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	0.000	**0.001**	0.000	.010/.014	.001/**.000**	.008/.014	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.000	**0.240**	0.119	0.000	0.000	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**	**.000/.000**

Note: 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), trained separately from Table II(a,b); best in **bold**, second underlined. Δt = mean step delay over detected events; DS-F1 = unweighted 7-dataset macro Buffered Point-F1. In (c), n = streams, K = channels, E = events, π = faulty-step fraction, $F1_{all} = 2\pi/(1+\pi)$ (trivial always-alarm Point-F1), Lam. = Laminar-JEPA, Lite = Laminar-JEPA-Lite. Unweighted Laminar-JEPA Δt reflects one late SWAN trigger (0.0002 recall; median Δt is 50.5 buffered / 45.8 causal, or 54.8 / 57.2 ex-SWAN). 95% stream-bootstrap CIs (n=19): Lite buffered FPR [.030, .103], F1 [.151, .335] and causal FPR [.006, .014], F1 [.055, .236]; Lam. buffered F1 [.150, .357] and causal FPR [.009, .018], F1 [.058, .257] (vs. Multi-Horizon 1D [.013, .097] and TimesNet causal FPR [.123, .375]).

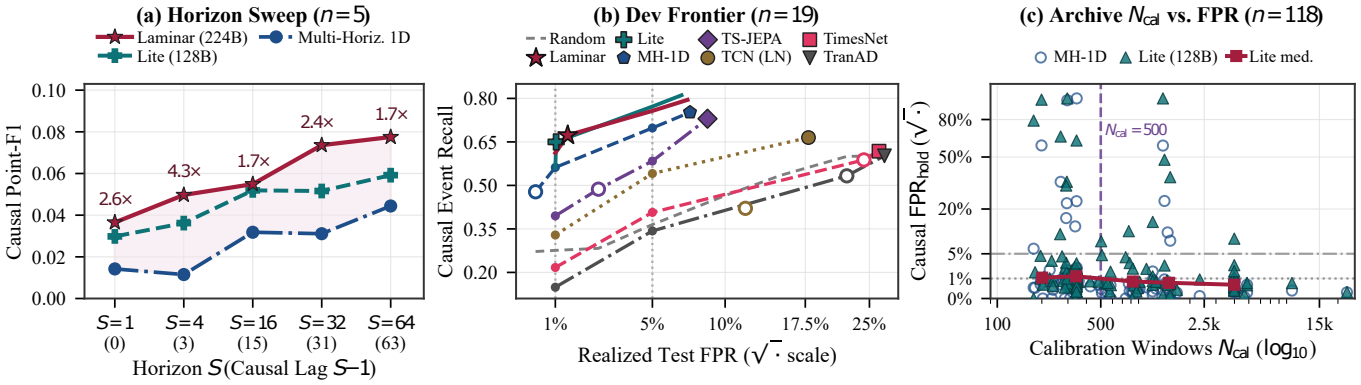


Fig. 4. Horizon scaling, operating frontier, and archive calibration-size effects. **(a)** Exploratory horizon sweep ($S \in \{1..64\}$, causal lag $S-1$, n=5 streams, nominal FPR $\leq$ 3.6% vs. 2.5% $\rightarrow$ 21.3% for TimesNet). **(b)** Design-stage causal ER vs. realized test FPR ($\sqrt{\cdot}$ scale, n=19 streams) across diagnostic budgets $f^* \in \{1\%, 5\%\}$ (dotted), autonomous points, and $q^* \leq 0.05$. **(c)** Primary archive calibration windows N_{cal} vs. causal held-out FPR_{hold} across n=118 untouched streams (Spearman $\rho = -0.32$, $p = 5.2 \times 10^{-4}$), marking N_{cal}=500 and 1%/5% budgets.

sequences [4], *Laminar-JEPA-Lite* and *Laminar-JEPA* reach 87.5% and 77.5% autonomous ER at 1.3% and 2.2% FPR versus 62.5% at 41.5% FPR for TimesNet.

E. Primary Held-Out Generalization, False-Alarm Tails, and Calibration Limits

Table VI and Fig. 4(c) establish three primary findings under the disjoint five-block protocol:

1) *Event-level gains, operating-point reconciliation, and calibration-length split (Table VI):* Across all 118 untouched streams (three seeds; on all 141 archive streams *Laminar-JEPA-Lite* reaches .076/.156 Point-F1 and .636/.658 ER vs. .075/.146 and .512/.573 for Multi-Horizon 1D), buffered and causal modes differ in how gains arise. Under *buffered* attribution, *Laminar-JEPA-Lite* raises ER from 59.0% to 67.4% (+8.4 points, cluster CI [+2.3%, +16.6%]) and Point-F1 from 0.143 to 0.156 ($p = 0.022$) while *lowering* buffered FPR_{hold} (19.5% $\rightarrow$ 14.3%, median 8.7% $\rightarrow$ 4.8%, $p = 0.0067$), and keeps a +11.0-point ER advantage at a post-hoc matched 5% test FPR (53.3% vs. 42.2%, $p = 1.2 \times 10^{-5}$, 7/9 datasets; Table VI(b)). Under live *causal* streaming, *Laminar-JEPA-Lite* raises autonomous ER from 52.5% to 63.8% (+11.3 points, cluster CI [+5.1%, +21.8%], 9/9 datasets, sign-flip $p = 0.0039$) and halves detection delay (70.8 vs. 133.5 steps; 66 vs. 136 in the rerun, $p = 4.8 \times 10^{-4}$), but pooled causal Point-F1 improves only modestly (0.076 $\rightarrow$ 0.079, Δ = +0.003, stream CI [−0.019, +0.023], cluster CI [−0.033, +0.060], 70/43 wins, $p = 0.014$) alongside higher mean causal FPR_{hold} (4.4% $\rightarrow$ 7.0%). Matching $\mathbf{X}_{cal}$ alarm rates (Table VI(b)), *Laminar-JEPA-Lite* still catches +12.4 points more causal events than rate-matched 1D (65.0% vs. 52.6%, $p = 1.7 \times 10^{-5}$), whereas at post-hoc equalized test FPR ($f^* \in \{1\%, 5\%\}$) causal ER matches

TABLE V
CONTEXTUAL VS. POINT ANOMALY DETECTION

(a) Model	NASA Contextual (n=5 seqs) Buf. PR	ER	FPR_{hold}	f^*=.01	f^*=.05	NASA Point (n=8) f^*=.01	f^*=.05
Laminar-JEPA	0.519	0.775	0.022	0.800	**0.960**	0.783	0.900
Laminar-JEPA-Lite	0.532	**0.875**	0.013	**0.840**	0.920	0.700	0.967
Multi-Horiz. 1D	**0.652**	0.750	**0.006**	0.820	0.920	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

(b) Model	19-Stream Contextual ($\lvert z\rvert \leq 5$) q^*=.05	f^*=.01	f^*=.05	19-Stream Point ($\lvert z\rvert > 5$) q^*=.05	f^*=.01	f^*=.05
Laminar-JEPA	**0.669**	0.448	**0.650**	0.835	0.644	0.795
Laminar-JEPA-Lite	0.655	0.376	0.567	**0.848**	0.581	**0.801**
Multi-Horiz. 1D	0.628	0.498	0.590	0.782	0.683	0.781
TS-JEPA	0.614	**0.520**	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Buf. PR = buffered PR-AUC; ER and FPR_{hold} = autonomous causal recall and held-out FPR; q^*=.05 = 95th-percentile $\mathbf{X}_{\text{cal}}$ ER; f^*=.01, .05 = diagnostic FPR_{test}-matched ER; best in **bold**, second underlined. (a) 10 SMAP/MSL streams by NASA taxonomy [4] (contextual: E-1, A-2, A-7, M-1, C-1 $[2100, 2210]$; point: A-1, P-3, D-3, D-14, D-15, C-1 $[550, 750]$). (b) n=19 non-GHL streams split into in-range ($\max |z| \leq 5\sigma$) vs. out-of-range ($\max |z| > 5\sigma$) causal events.

TABLE VI
DISJOINT FIVE-BLOCK UNTOUCHED ARCHIVE BENCHMARK (n=118)

(a) Archive (n=118)	Pt-F1 ↑ Caus.	Buf.	ER ↑ Caus.	Buf.	FPR_{hold} Caus.	Buf.	Caus. FPR Tail Med.	p90	p95	≤5%
TS-JEPA (1D)	.057	.134	.414	.459	**.042**	**.110**	**.004**	.070	**.209**	88.1
Multi-Horiz. 1D	.076	.143	.525	.590	.044	.195	**.004**	**.068**	.237	**89.0**
Laminar-JEPA (224B)	.072	.144	.609	.619	.068§	.190	.013	.076	.383	86.4
Laminar-Lite (128B)	**.079**†	**.156**‡	**.638**¶	**.674**	.070	.143	.008	.106	.502	87.3
Adequate (N_{cal}≥500, n=59) vs. short (N_{cal}<500, n=59) histories										
MH-1D, $N_{\text{cal}}\geq 500$	.040	.096	.368	.457	**.020**	.133	**.002**	**.009**	**.086**	**93.2**
Lite, $N_{\text{cal}}\geq 500$	**.079**	**.127**	**.528**	**.577**	.047	**.099**	.006	.084	.169	88.1
MH-1D, $N_{\text{cal}}<500$	**.112**	**.191**	.684	.727	**.069**	.258	**.007**	**.175**	**.364**	84.7
Lite, $N_{\text{cal}}<500$	.080	.186	**.750**	**.773**	.093	**.188**	.012	.323	.669	**86.4**

(b) Rerun (n=118)	Causal Pt-F1	ER	$\text{ER}_{.05}$	Mean	Med.	Δt	Buffered Pt-F1	ER	$\text{ER}_{.05}$
Multi-Horizon 1D (SPOT)	.074	.502	.752	**.046**	**.003**	136	.145	.586	.422
at Lite's cal. rate	**.085**	.526	.752	.050	.005	110	.146	.576	.422
Laminar-JEPA-Lite	.078	**.650**	.726	.071	.010	**66**	.154	.661	**.533**
at 1D's cal. rate	.061	.563	.726	.062	.004	95	**.157**	**.698**	**.533**
Laminar-JEPA, 224 dials	.074	.646	.767	.068	.013	70	.143	.643	.498
Laminar-JEPA, 160 dials	.073	.616	.736	.064	.009	88	.140	.634	.529
Lite res., raw top-k	.069	.581	.726	.073	.006	78	–	–	–
Lite res., max-z	.063	.588	**.797**	.067	.006	69	–	–	–
Lite res., Fisher	.077	.611	.722	.080	.010	79	–	–	–

(a) Three seeds; best **bold**, second underlined; MH-1D = Multi-Horizon JEPA (1D), Lite = Laminar-JEPA-Lite; ≤5% = % of streams with causal $\text{FPR}_{\text{hold}}\leq 5\%$. PR-AUC (caus./buf.): TS-JEPA .155/.207, 1D .162/.230, 224B .143/.211, Lite .145/.224; unquenched 224B: .075/.141 F1, .644/.640 ER, .069/.190 FPR_{hold}. †Lite vs. 1D causal F1 Δ=+0.003 (stream CI $[-0.019, +0.023]$, cluster CI $[-0.033, +0.060]$; 70/43 wins, p=0.014); vs. TS-JEPA +0.022 (cluster CI $[+0.005, +0.052]$). ‡Buf. F1 Δ=+0.014 (p=0.022). ¶Causal ER Δ=+0.113 (cluster CI $[+0.051, +0.218]$, 9/9 sets) at higher causal FPR_{hold} (+0.025, cluster CI $[+0.006, +0.055]$). §Disjoint quench: FPR_{hold} Δ=−0.0007 (p=0.55), ER −0.035. (b) Independent rerun on identical splits; Mean/Med. = causal FPR_{hold}; $\text{ER}_{.05}$ = diagnostic recall at 5% test FPR; "at cal. rate" matches $\mathbf{X}_{\text{cal}}$ alarm rates (Stouffer .057 F1, .558 ER; ℓ_2^2 conf. .071, .556; BH-FDR .024, .146).

1D (72.6% vs. 75.2% at 5%, $p = 0.20$). Splitting at the development-suite threshold $N_{\text{cal}} = 500$ explains the modest pooled Point-F1: on adequately calibrated streams ($N_{\text{cal}} \geq 500$, $n = 59$), *Laminar-JEPA-Lite* doubles causal Point-F1 over 1D (0.079 vs. 0.040, $\Delta = +0.039$, CI $[+0.016, +0.064]$, $p = 2.7 \times 10^{-5}$; +16.0 ER points) and still wins without Exathlon ($n = 43$: 0.051 vs. 0.031, $p = 0.0050$, mean $\text{FPR}_{\text{hold}} = 1.56\%$), whereas on short histories ($N_{\text{cal}} < 500$, $n = 59$) 1D leads causal Point-F1 (0.112 vs. 0.080, $p = 0.31$; explaining why pooled non-Exathlon $n = 101$ favors 1D on causal F1, 0.073 vs. 0.060, while *Laminar-JEPA-Lite* leads buffered F1 0.148 vs. 0.139 and causal ER 58.8% vs. 50.6%). Disjoint quenching barely changes causal FPR_{hold} (6.90% → 6.83%, $p = 0.55$) while costing 3.5 ER points.

2) *Held-out false-alarm tail distribution, operational budgets, and autocorrelation (Table VI(a), Fig. 4(c)):* Unlike the ≈ 1% development results, archive causal FPR_{hold} is strongly right-skewed across $n = 118$ streams. Mean, median, 90th, and 95th percentiles are $(4.2\%, 0.38\%, 7.0\%, 20.9\%)$ for TS-JEPA, $(4.4\%, 0.39\%, 6.8\%, 23.7\%)$ for Multi-Horizon 1D, $(6.8\%, 1.26\%, 7.6\%, 38.3\%)$ for *Laminar-JEPA*, and $(7.0\%, 0.78\%, 10.6\%, 50.2\%)$ for *Laminar-JEPA-Lite*. Stream counts exceeding $(1\%, 2\%, 5\%, 10\%)$ causal FPR_{hold} are $(29, 21, 14, 10)$ for TS-JEPA, $(23, 16, 13, 11)$ for Multi-Horizon 1D, $(67, 41, 16, 12)$ for *Laminar-JEPA*, and $(54, 33, 15, 13)$ for *Laminar-JEPA-Lite*; thus 54.2% of *Laminar-JEPA-Lite* streams meet a ≤ 1% budget (66.1% when $N_{\text{cal}} \geq 500$) and 87.3% meet a ≤ 5% budget (88.1% when $N_{\text{cal}} \geq 500$). Calibration size correlates inversely with FPR_{hold} (Spearman $\rho = -0.32$, $p = 5.2 \times 10^{-4}$): $N_{\text{cal}} < 250$ ($n = 12$) averages 16.0% causal FPR_{hold} versus 1.05% (median 0.48%) for $N_{\text{cal}} \geq 2{,}500$ ($n = 22$). Autocorrelation drives this finite-sample variance: on the 19 development streams, median Sokal τ_{int} is 2.51 (z_j), 1.88 (N_{edge}), 5.58 (N_{224}, max 206.0), and 6.75 (1D), leaving median N_{eff} of $(154.6, 67.1, 195.9)$ at median $N_{\text{cal}} = 408$ and only $(2.32, 0.34, 0.20)$ expected tail exceedances (12.7%–25.0% block-bootstrap CI half-widths). On short histories ($T_{\text{nom}} < 2{,}500$, $n = 18$), two-fold cross-fitting cuts causal FPR_{test} from 15.4% to 3.8% at 69.9% ER.

3) *Regime drift vs. tail misfit:* Comparing chronological (FPR_{chr}), $L = 64$ block-permuted (FPR_{blk}), and point-wise i.i.d. ($\text{FPR}_{\text{pt}} \leq 0.35\%$) splits links excess FPR primarily to inter-block drift without ruling out tail misfit ($\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$ accounts for 71.3% of excess causal FPR in TS-JEPA and 78.3%–81.8% in reconstructors vs. 18.7% with centering). Within-checkpoint traces (n=6, $K \in \{9, 25, 55\}$) show tap fusion and centering lift anomaly divergence from 0.06–0.16 to 0.70–0.94 and cut mean coordinate D_{KS} from 0.140 to 0.096.

VIII. CONCLUSION AND FUTURE WORK

We studied multi-horizon circuit-breaker panels (*Laminar-JEPA-Lite* for operational deployment; 224-dial *Laminar-JEPA* as an ablation reference) for abrupt and intermediate-onset telemetry faults across 33 exploratory design-stage streams and 118 primary untouched archive streams:

1) *Event coverage, buffered accuracy, and operating points:* Under the five-block archive protocol ($n = 118$), *Laminar-JEPA-Lite* raises buffered ER (59.0% → 67.4%) and Point-F1 (0.143 → 0.156) at lower held-out FPR (19.5% → 14.3%), and raises live causal ER (52.5% → 63.8%) across all nine datasets ($p = 0.0039$), halves detection delay, and beats system-matched coordinate aggregators at 35%–52% lower parameter cost and 2.43× speedup over *Laminar-JEPA*. Pooled causal Point-F1 rises modestly (0.076 → 0.079), doubling on adequately calibrated streams ($N_{\text{cal}} \geq 500$: 0.040 → 0.079, $p = 2.7 \times 10^{-5}$), while weight quenching yields no reliable benefit.
2) *Contributing factors and latency:* Design-stage matched-backbone controls indicate that LayerNorm and centering

contribute to lighter tails and lower drift under these protocols, and an exploratory five-stream sweep preserves $2.1\times$ –$4.3\times$ causal Point-F1 gains at $S \in \{1, 4\}$.

3) *False-alarm tails, single-sensor, and ramp limits:* Archive causal FPR_{hold} has a low median (0.78%; 87.3% of streams $\leq 5\%$) but a 7.0% mean driven by short autocorrelated calibration blocks (two-fold cross-fitting cuts short-stream FPR_{test} from 15.4% to 3.8%). An optional channel branch helps synthetic single-sensor faults only at elevated FPR, and slow ramps ($L \geq 4C$, including GHL at 0.000 Point-F1) and natural sensor failures remain open priorities.

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